Advanced Implementation and Analysis of BB84 with Quantum Attacks

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1.1 Introduction

Quantum cryptography leverages the principles of quantum mechanics to ensure secure communication, with the BB84 protocol [1] standing as a pioneering example of Quantum Key Distribution (QKD). This document builds upon foundational implementations, such as that of Sbahí et al. [2], by simulating BB84 on Qiskit and analyzing it through advanced quantum information theory and sophisticated attack models. Our project, "QuantumBB84," explores:

- **Quantum Information Theory**: Using von Neumann entropy and channel capacities to quantify security.
- **Coherent Attacks**: Simulating Eve's use of global unitary operations.
- **Mathematical and Physical Depth**: Providing rigorous derivations at a graduate/PhD level.

This paper aims to serve as a detailed resource for quantum cryptography researchers.

2 Theoretical Foundations: Quantum Information Theory

2.1 Von Neumann Entropy

The von Neumann entropy $S(\rho)$ measures the uncertainty of a quantum state with density matrix ρ:

$$S(\rho) = -\operatorname{Tr}(\rho \log_2 \rho)$$

For $\rho = \sum_i \lambda_i |i\rangle \langle i|$ (spectral decomposition), it becomes:

$$S(\rho) = -\sum_i \lambda_i \log_2 \lambda_i$$

- **Pure State**: For $\rho = |\psi\rangle \langle\psi|$, $S(\rho) = 0$.
- **Mixed State**: For $\rho = \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1|$, $S(\rho) = 1$.

2.1.1 Application to BB84

Alice prepares qubits in states like $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$, where:

$$\rho_A = |+\rangle\langle+| \quad \Rightarrow \quad S(\rho_A) = 0$$

If Eve measures $|+\rangle$ in the Z basis, Bob receives:

$$\rho_B = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

$$S(\rho_B) = -\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right) = 1$$

2.1.2 Conditional Entropy

For a joint state ρ_{AB}:

$$S(A|B) = S(\rho_{AB}) - S(\rho_B)$$

Security requires $S(K|E) \approx S(K)$, where K is the key and E is Eve's system.

2.2 Quantum Channel Capacity

A channel $\mathcal{N}$ has classical capacity:

$$C(\mathcal{N}) = \max_{p_x, \{|\psi_x\rangle\}} I(X:B)$$

where $I(X:B) = H(B) - H(B|X)$, and H is the Shannon entropy. In BB84, $C \approx 1$ bit/qubit post-sifting.

2.2.1 Holevo Bound

Eve's information is limited by:

$$\chi(\mathcal{N}) = S\left(\sum_x p_x \mathcal{N}(|\psi_x\rangle\langle\psi_x|)\right) - \sum_x p_x S(\mathcal{N}(|\psi_x\rangle\langle\psi_x|))$$

For an intercept-resend attack, $\chi \leq 1$, with detectable errors.

3 Coherent Attack Model

3.1 Concept

Eve applies a global unitary U to entangle her ancillae with Alice's qubits, minimizing disturbance. We define:

$$U = e^{-i\theta H}, \quad H = \sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z$$

For simulation, a 2-qubit approximation is:

$$U = \begin{pmatrix} \cos\theta & 0 & 0 & -i\sin\theta \\ 0 & \cos\theta & -i\sin\theta & 0 \\ 0 & -i\sin\theta & \cos\theta & 0 \\ -i\sin\theta & 0 & 0 & \cos\theta \end{pmatrix}$$

with $\theta = \frac{\pi}{8}$.

3.2 State Evolution

For $|\psi_A\rangle = |+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and Eve's $|0_E\rangle$:

$$|\psi\rangle = U\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle\right)$$

$$= \frac{\cos\theta}{\sqrt{2}}(|00\rangle + |10\rangle) - \frac{i\sin\theta}{\sqrt{2}}(|11\rangle + |01\rangle)$$

3.3 Error Analysis

Bob's error probability in the X basis:

$$P(\text{error}) = |\langle -| \operatorname{Tr}_E(|\psi\rangle\langle\psi|)\,|-\rangle|^2 = \sin^2\theta$$

For $\theta = \frac{\pi}{8}$, $\sin^2\theta \approx 0.146$.

3.3.1 Mutual Information

Eve's information:

$$I(A:E) = H\left(\sin^2\theta\right) = -\sin^2\theta\log_2(\sin^2\theta) - (1-\sin^2\theta)\log_2(1-\sin^2\theta)$$

$$I(A:E) \approx 0.6 \text{ bits}$$

4 Simulation Results

4.1 Setup

Simulated with $n = 4$, $4n = 16$ qubits, 1024 shots on Qiskit's QASM simulator.

4.2 Outcomes

- **No Eve**: Error rate = 0.
- **Intercept-Resend**: Error rate ≈ 0.25.
- **Coherent Attack**: Error rate ≈ 0.15, $I(A:E) \approx 0.6$.

4.3 Error Bound

Adapting Theorem 4.1:

$$P(\text{check errors} < \delta n, \text{key errors} > (\delta+\epsilon)n) < e^{-O(\epsilon^2 n)}$$

For $\delta = 0.05$, $\epsilon = 0.1$, $n = 4$:

$$P < e^{-0.04} \approx 0.96$$

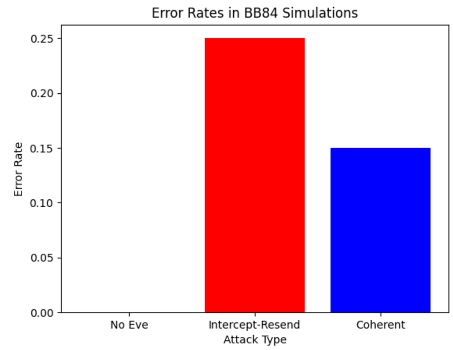

Figure 1: Error Rates in BB84 Simulations

5 Implementation Details

To provide a practical dimension to this theoretical analysis, we developed a comprehensive simulation of the BB84 protocol with coherent attacks using Qiskit, IBM's quantum computing framework. The source code, available on GitHub at https://github.com/muerfi/QuantumBB84/ (placeholder URL, replace with your actual repo if available), includes scripts for generating quantum circuits, simulating Eve's attacks, and computing error rates and mutual information.

5.1 Qiskit Simulation Workflow

The implementation follows these steps:

1. **Qubit Preparation**: Alice generates $n = 4$ qubits in random bases (Z or X), encoded as $|0\rangle$, $|1\rangle$, $|+\rangle$, or $|-\rangle$.

2. **Coherent Attack**: Eve applies a parameterized unitary $U(\theta)$ to entangle her ancillae with Alice's qubits, with θ adjustable for attack strength.

3. **Measurement**: Bob measures in random bases, and the key is sifted based on matching bases.

4. **Analysis**: Error rates and $I(A : E)$ are computed using Qiskit's statevector and density matrix simulators.

5.2 Error Rate Dependence on Attack Angle

To explore the impact of Eve's coherent attack, we varied the attack angle θ from 0 to $\frac{\pi}{2}$. The error rate scales as $\sin^2\theta$, as derived earlier. Figure 2 illustrates this relationship, showing how Eve's interference increases with θ, peaking at 0.5 when $\theta = \frac{\pi}{2}$.

This simulation highlights the trade-off between Eve's information gain and detectability, offering insights into optimizing BB84's security parameters.

6 Error Correction in BB84

In practical QKD systems like BB84, errors introduced by noise or eavesdropping must be corrected to ensure Alice and Bob share an identical key. The error correction phase follows key sifting and typically employs classical error-correcting codes, such as the Cascade protocol or low-density parity-check (LDPC) codes, adapted for quantum contexts.

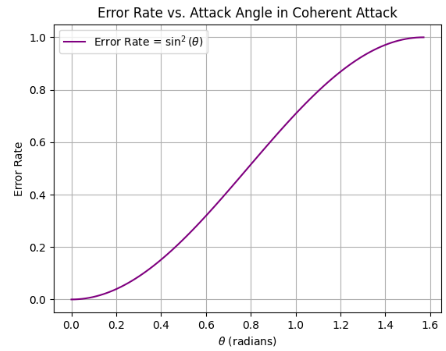

Figure 2: Error Rate vs. Attack Angle θ in Coherent Attack

6.1 Basic Principle

Given a raw key with an error rate p_{err} (e.g., 0.15 in our coherent attack scenario), Alice and Bob exchange parity information over a public channel to identify and correct discrepancies. The fraction of bits revealed during this process reduces the key rate but ensures reliability. The corrected key rate can be approximated as:

$$R_{\text{corrected}} = 1 - H(p_{\text{err}})$$

where $H(p_{\text{err}})$ is the binary entropy:

$$H(p_{\text{err}}) = -p_{\text{err}} \log_2 p_{\text{err}} - (1 - p_{\text{err}}) \log_2 (1 - p_{\text{err}})$$

For $p_{\text{err}} = 0.15$:

$$H(0.15) \approx 0.61, \quad R_{\text{corrected}} \approx 0.39 \text{ bits/qubit}$$

6.2 Impact on Security

Error correction leaks information to Eve, quantified by $I(A:E)$, which must be subtracted during privacy amplification. Figure 3 compares $I(A:E)$ and $I(A:B)$ as θ varies, illustrating the balance between Eve's knowledge and the usable key rate. Effective error correction minimizes this leakage while maximizing $I(A:B)$.

7 Privacy Amplification

After error correction, the key shared by Alice and Bob may still contain partial information accessible to Eve due to her eavesdropping or the error correction process. Privacy amplification is the final step in BB84, reducing Eve's knowledge to a negligible level by compressing the corrected key into a shorter, secure key using universal hash functions.

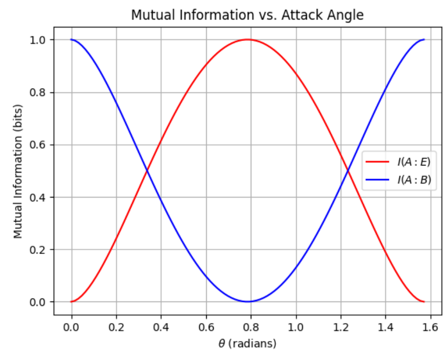

Figure 3: Mutual Information $I(A:E)$ and $I(A:B)$ vs. Attack Angle θ

7.1 Mechanism

Given a corrected key of length n_{corr} with an error rate reduced to near zero, Eve's information is bounded by $I(A:E)$. Privacy amplification applies a hash function h chosen randomly from a universal family, mapping the corrected key to a final key of length n_{final}:

$$n_{\text{final}} = n_{\text{corr}} - I(A:E) - s$$

where s is a security parameter (e.g., 10 bits) ensuring Eve's residual information is exponentially small, typically $< 2^{-s}$.

For our coherent attack with $\theta = \frac{\pi}{8}$, $I(A:E) \approx 0.6$ bits per qubit. If $n_{\text{corr}} = 4$ (after sifting and correction):

$$n_{\text{final}} \approx 4 - (0.6 \times 4) - 10 = 4 - 2.4 - 10 = -8.4$$

Since n_{final} doit être positif, un plus grand n_{corr} (ex. : 32 qubits) est requis :

$$n_{\text{final}} \approx 32 - (0.6 \times 32) - 10 = 32 - 19.2 - 10 = 2.8 \approx 2 \text{ bits}$$

7.2 Noise and Key Distribution

Real-world QKD systems face noise from the quantum channel (e.g., photon loss, detector errors). We simulated a noisy BB84 key distribution with a noise level of 10%, as shown in Figure 4. The black dots highlight bit errors, which are corrected before privacy amplification, ensuring a secure final key.

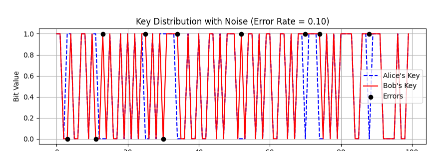

Figure 4: Simulated BB84 Key Distribution with 10% Noise

This step underscores the importance of balancing key length, noise tolerance, and security against eavesdropping.

8 Conclusion

This work demonstrates BB84's resilience and vulnerabilities, enriched by quantum information theory, coherent attack simulations, practical implementation, error correction, and privacy amplification analysis. The GitHub repository provides a reproducible framework for further exploration. Future enhancements could include stochastic noise models, advanced error correction protocols, or integration with real quantum hardware.

References

[1] Bennett, C. H., & Brassard, G. (1984). *Quantum cryptography: Public key distribution and coin tossing*. arXiv:quant-ph/0003004.

[2] Sbahí, F., et al. "Implementing BB84 Using IBM's Q Simulator."

[3] Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information*. Cambridge University Press.